Statistical Learning Project Report

Done by Chaima Ben Fredj, ID: 56374A

Introduction

This project applies both supervised and unsupervised machine learning techniques to two distinct datasets: a heart disease dataset analyzed through supervised models (Logistic Regression, SVM, Decision Trees, Random Forest) to predict cardiovascular risk, and a Country dataset explored via unsupervised methods (PCA, K-means, hierarchical clustering) to identify natural socioeconomic groupings without predefined labels. Together, these approaches demonstrate machine learning's capabilities in predicting healthcare diagnostics and exploring socioeconomic patterns.

Unsupervised Learning

Understanding world development trends is necessary for policymakers, international institutions, and economists in order to create informed resource allocation and targeted intervention strategies. After exploring our data and preprocessing it, we applied unsupervised learning techniques - specifically K-means clustering, hierarchical clustering, and Principal Component Analysis (PCA) - to identify patterns automatically in unlabeled country data. These models revealed natural groupings of countries by economic and health factors, in order to classify countries by development categories without pre-designated labels.

Dataset

In this part of our project, we use unsupervised learning to explore socioeconomic and health variables of countries in search of natural groupings reflecting various levels of development without pre-specified labels.

Our dataset selected from kaggle, contains 166 observations and 10 variables which we can classify as the following:

Economic metrics: GDP per capita, income, inflation, exports/imports

Health metrics: Child mortality, life expectancy, fertility rate, health expenditure

Variable Name	Type	Description
Country	Character	Name of the country
child_mort	Numerical	Death of children under 5 years of age per 1000 live births
exports	Numerical	Exports of goods and services per capita. Given as %age of the GDP per capita
health	Numerical	Total health spending per capita. Given as %age of GDP per capita
imports	Numerical	Imports of goods and services per capita. Given as %age of the GDP per capita
Income	Integer	Net income per person
Inflation	Numerical	The measurement of the annual growth rate of the Total GDP
life_expec	Numerical	The average number of years a new born child would live if the current mortality patterns are to remain the same
total_fer	Numerical	The number of children that would be born to each woman if the current age-fertility rates remain the same.
gdpp	Integer	The GDP per capita. Calculated as the Total GDP divided by the total population.

Table 1: Dataset Variables

A. Exploratory Data Analysis (EDA)

I. Univariate Analysis

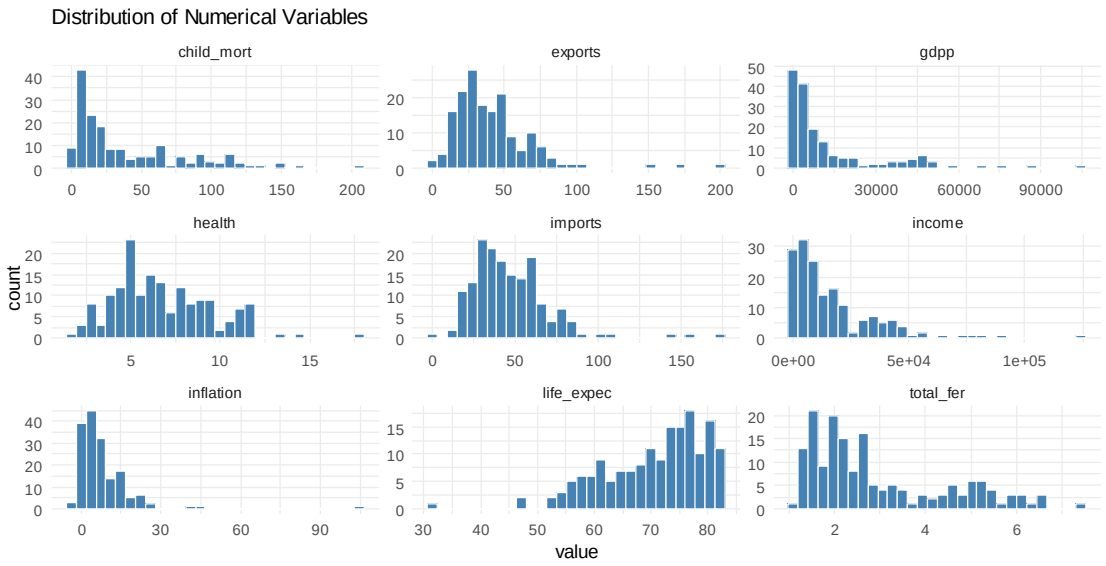


Figure 1: Numercial variables distribution

From the distribution of our variables we can observe that:

Economic indicators (GDP, income) were right-skewed, with most countries at lower economic status

Health indicators portrayed spectacular differences: child mortality peaked in some nations (0-200/1,000 births)

Trade activity (exports/imports) was limited for most countries

These distributions highlight global inequalities, highlighting the need for clustering to group countries by development stage.

II. Bivariate Analysis

Scatter Plots

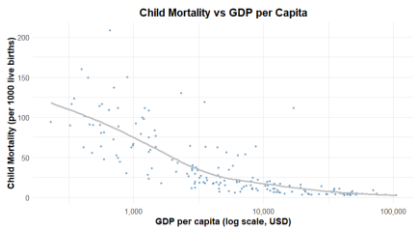


Figure 2: Child Mortality vs GDPP

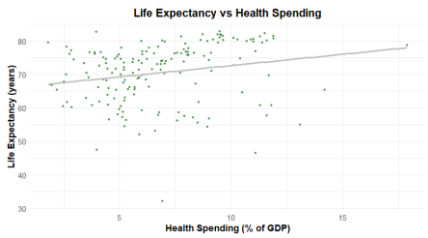


Figure 3: Life Expectancy vs Health

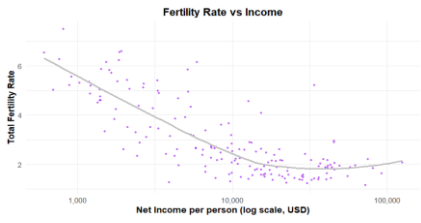


Figure 4: Fertility rate vs Income

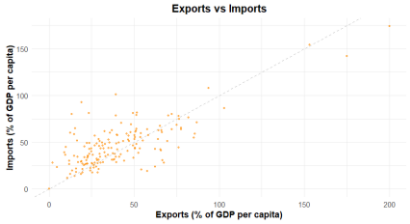


Figure 5: Exports vs Imports



Figure 6: Life Expectancy vs Child mortality

From these scatterplots we can interpret:

strong negative relationship between GDP per capita and child mortality.

Positive linear relationship between health spending (% of GDP) and life expectancy.

Strong negative relationship between net income per person and fertility rate.

Positive relationship between exports and imports.

Strong negative relationship between life expectancy and child mortality.

Bar Plots (grouped by indicators)

1. Countries by GDP per Capita

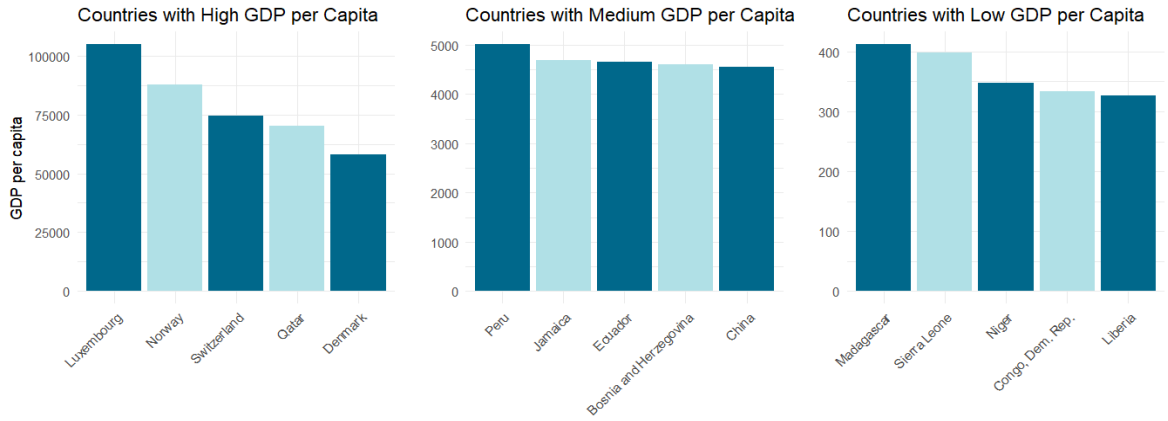


Figure 7: Countries vs GDPP

Gdpp is a key economic indicator that measures the average economic output per individual in a country. from these bar plots we can suggest that:

the *Top 5* countries represent highly developed economies and high standards of living.

Middle 5 countries: Represent a moderate level of development

Bottom 5 countries: countries suffering from low productivity, economic challenges, and weaker infrastructures.

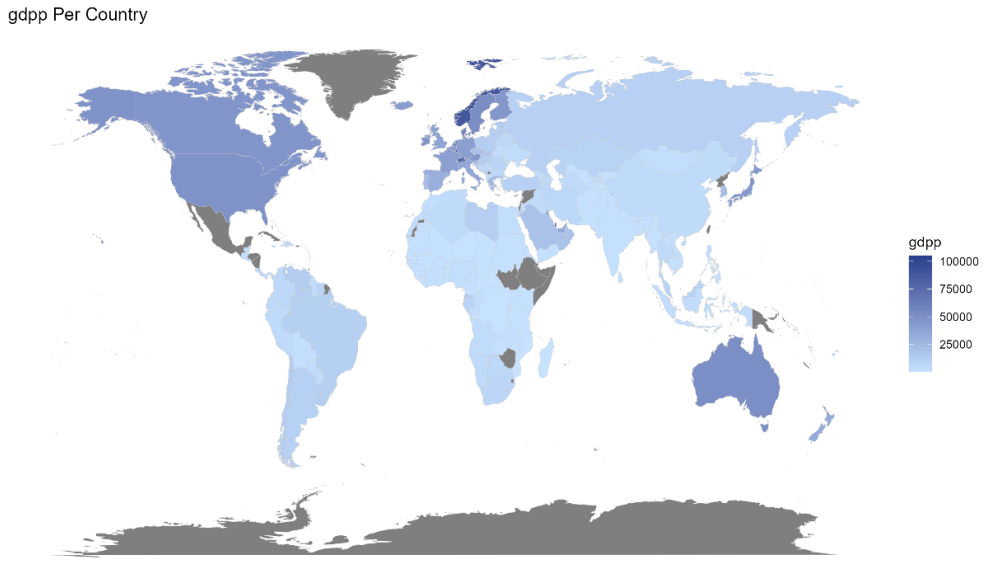


Figure 8: GDPP per country

2. Countries by Inflation Rate

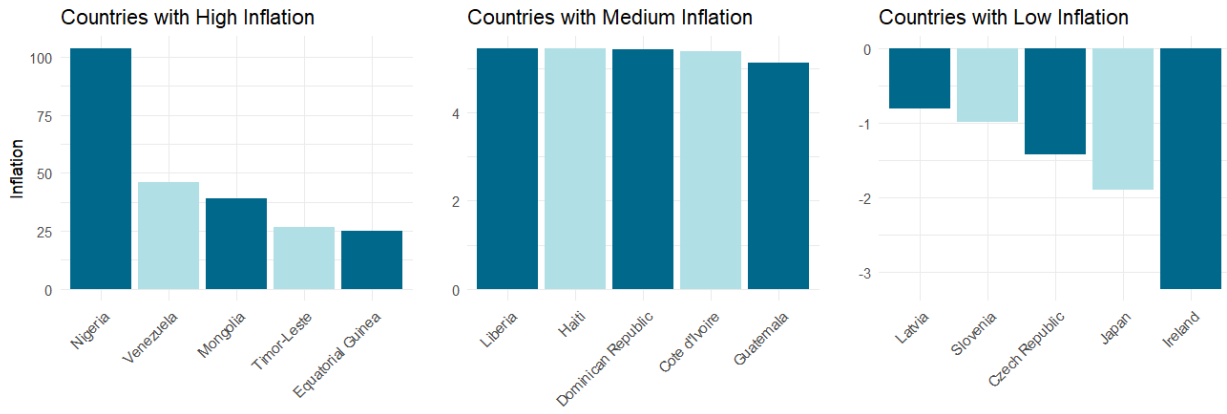


Figure 9: Country vs Inflation Rate

Top 5: Countries suffering from very high inflation, often linked with economic instability, currency devaluation, or political crises.

Middle 5: Moderate, often healthy inflation rates.

Bottom 5: Extremely low or even deflationary countries, which can also be unhealthy for an economy if it indicates stagnation.

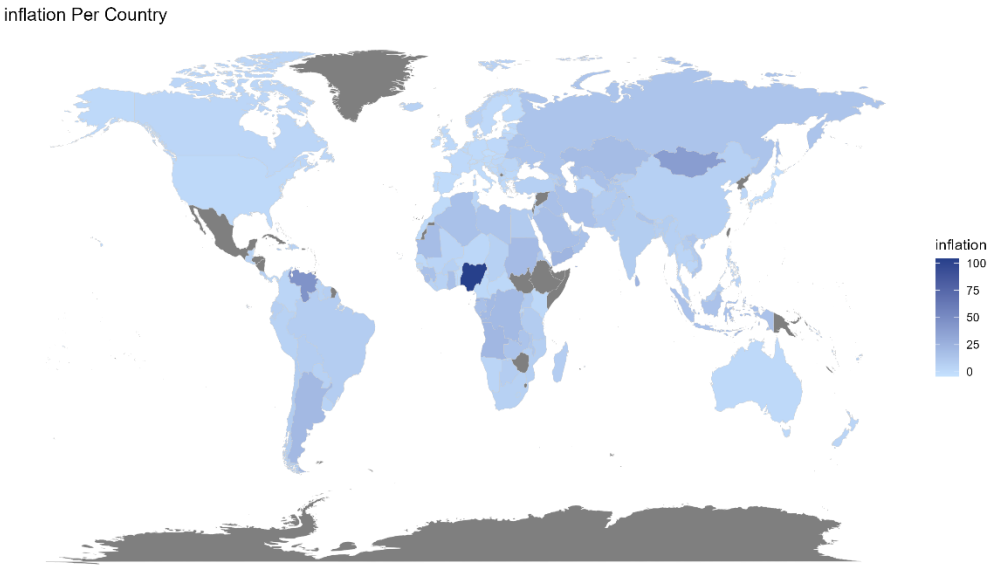


Figure 10: inflation rate per Country

3. Countries by Child Mortality

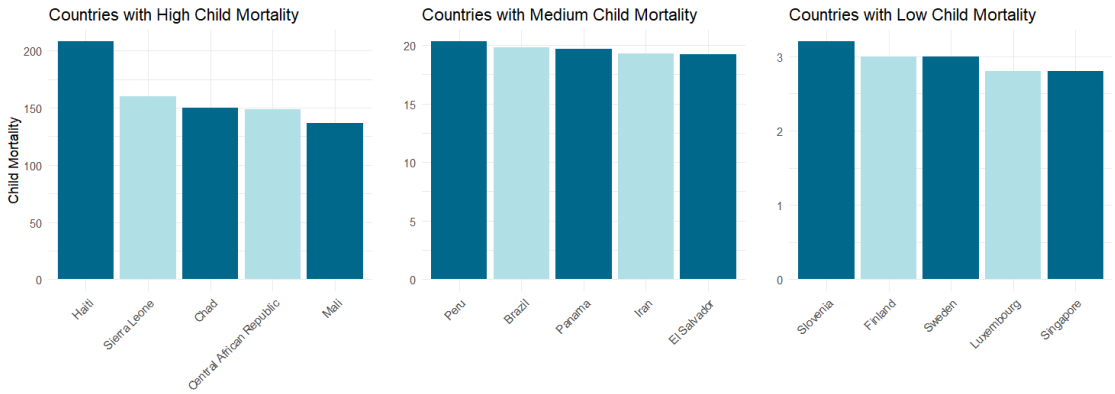


Figure 11: Country vs Child Mortality

Top 5: Very high child mortality rates, indicating poor healthcare systems, or malnutrition.

Middle 5: Average mortality, but still can be serious in global health standards.

Bottom 5: Very low mortality, generally developed countries with excellent healthcare.

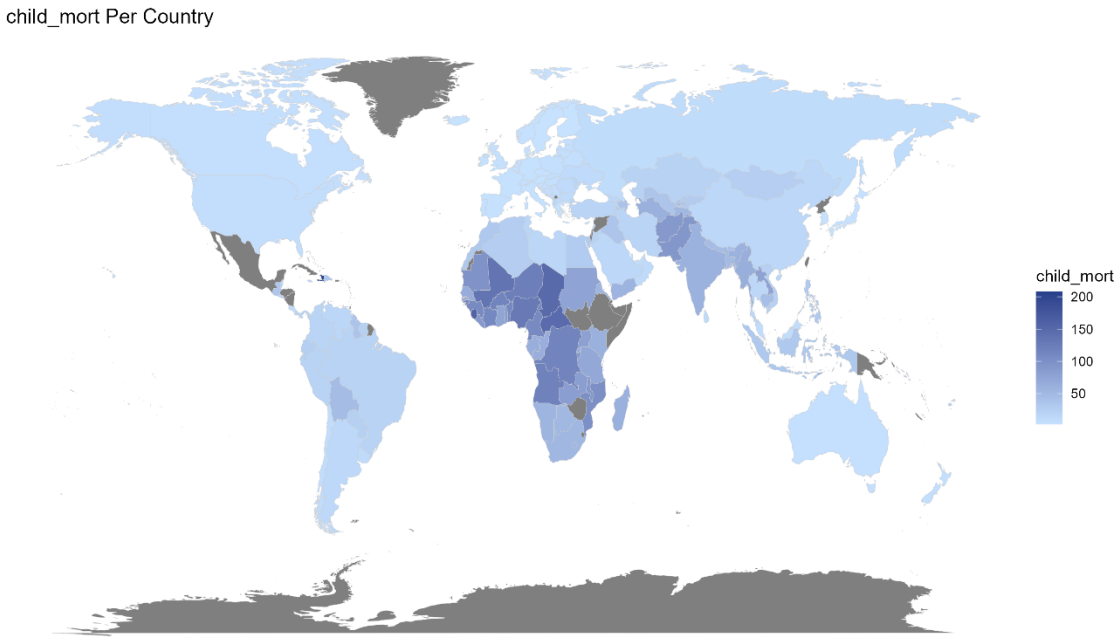


Figure12: Child Mortality per Country

III. Multivariate Analysis

To better understand the relationship between our variables, we generated the following Heatmap.

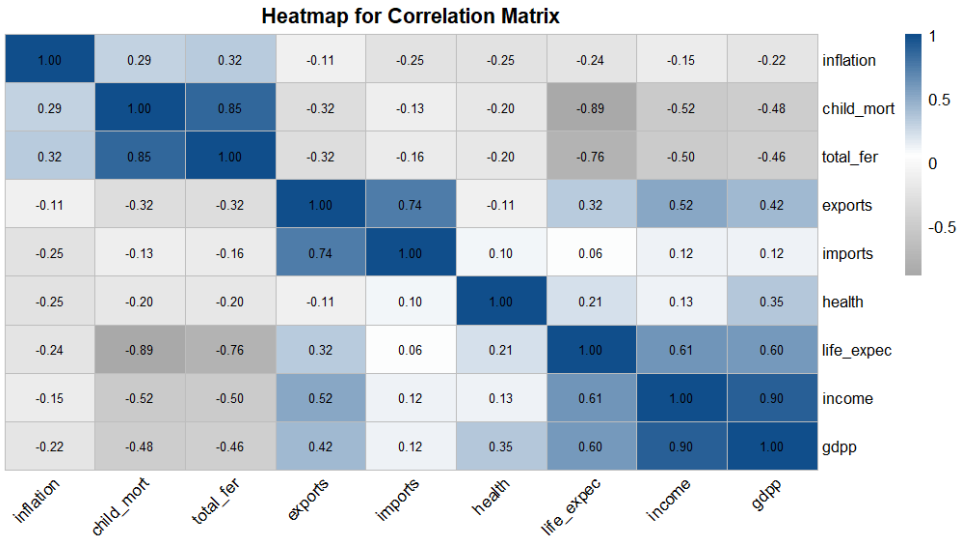


Figure13: Heatmap

The Heatmap shows a high positive correlation between income and gdpp (0.9), child mortality and total fertility (0.85), exports and imports (0.74)

However, child mortality is negatively correlated with most of the economic variables (except inflation as high inflation also means less stable economies), indicating that better economic metrics reveal lower child mortality.

Additionally, exports and imports showed a high positive correlation, implying that economies engaged in high levels of exports are also active in imports. However, Inflation shows weak correlations with economic variables.

B. Data Preprocessing

Prior to generating choropleth maps, data cleaning steps were taken to ensure compatibility between the country names in our dataset and the external mapping data (map_data("world")). Several country names were manually corrected, and a new mapping variable (map_name) was created to facilitate a successful join between datasets.

In addition, to prepare our dataset for clustering, first we removed the income variable since, as we have seen from our EDA, it's highly correlated with gdpp, then we applied standardization, we used z-score normalization to all our numeric variables and of course while excluding the country variable.

Principle Component Analysis (PCA)

After scaling, we applied PCA to the standardized data to reduce the dimensionality while retaining most of the dataset's variance. We retained the first three principal components as their cumulative variance is approximately 77.8%

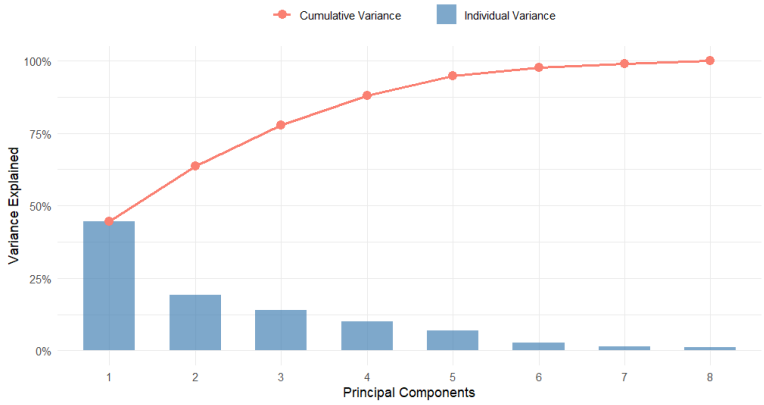


Figure 14: Explained Variance

After knowing that we will select three PCs, we will figure out which factors affect each principal components more, let's have a look at these PCA variable plots:

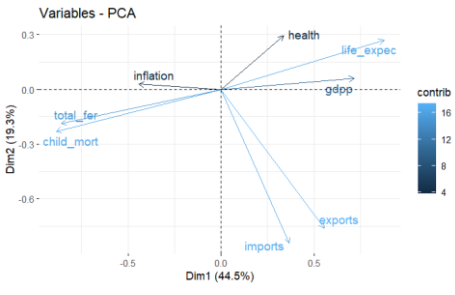


Figure 15: PC1 vs PC2

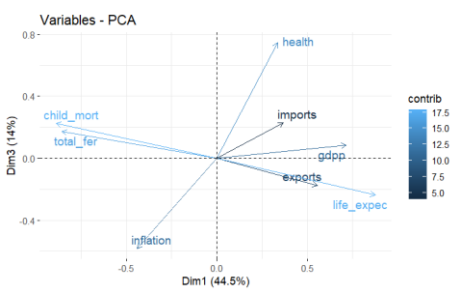


Figure 16: PC1 vs PC3

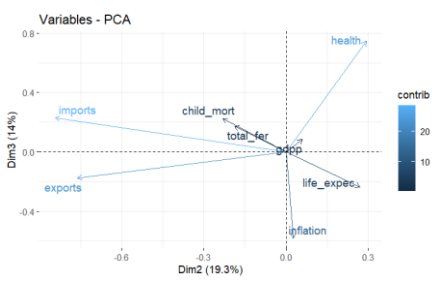


Figure 17: PC3 vs PC2

PCA Loadings (Top 3 Components)

Variables	**PC1**	**PC2**	**PC3**
child_mort	-0.47	-0.186	0.212
exports	0.294	-0.613	-0.165
health	0.179	0.236	0.704
imports	0.195	-0.679	0.215
inflation	-0.234	0.022	-0.55
life_expec	0.465	0.218	-0.224
total_fer	-0.455	-0.152	0.164
gdpp	0.38	0.048	0.08

Table 2: PCA Loadings

We can interpret that: *PC1 (44.5%)* primarily separates countries based on economic development metrics, with high positive loadings for GDP per capita, exports, and life expectancy, and high negative loadings for child mortality and fertility rates.

PC2 (19.3%) separates countries based on trade operations (exports and imports) vs health-related variables.

PC3 (14%) captures the contrast between health expenditure and inflation levels.
The PCA loading plots (variable factor maps) support these findings by visualizing the strong relationships and oppositions among the variables across different dimensions.

C. Models Implementation

Kmeans Clustering

K-means clustering helps us group similar countries together based on their development indicators. Imagine sorting nations by economic and health factors - the algorithm does this automatically, finding natural groupings without us having to define categories first. By analyzing GDP, child mortality rates, life expectancy and other key metrics together, it reveals patterns we might miss with traditional classification methods.

Finding k with the elbow method:

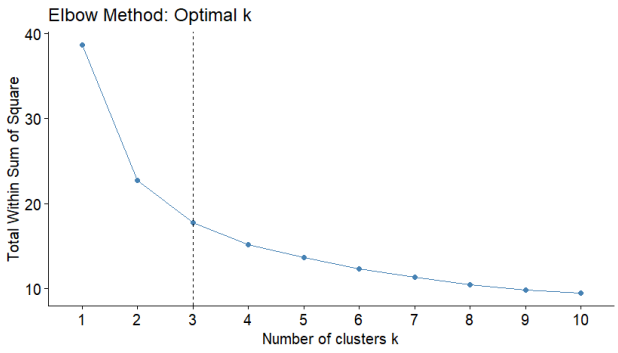


Figure 18 : Elbow Method

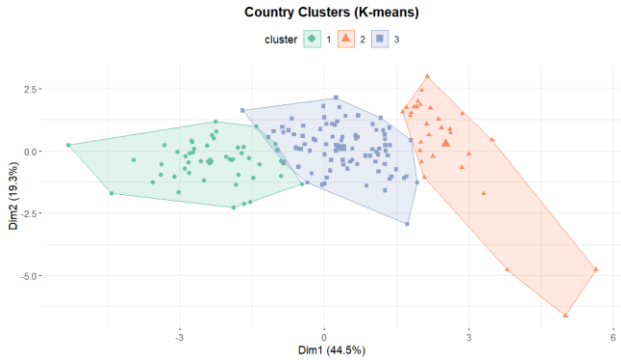


Figure 19 : kmeans clusters (k=3)

We used the elbow method to determine the ideal number of clusters by analyzing how total within-cluster variation (WSS) changes as clusters increase. According to the elbow method (see the plot above) the optimal number of clusters is 3.

Since we don't know which cluster corresponds to which group of countries we will mainly evaluate each cluster based on child mortality and gdpp since high child mortality rates and low gdpp are associated with least developed countries

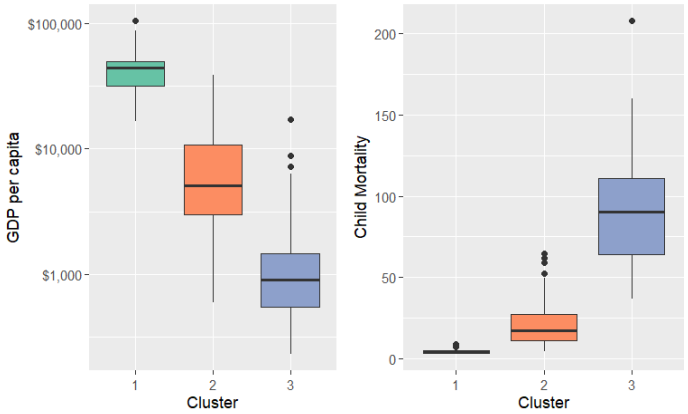


Figure 20: Clusters Boxplots

According to these boxplots we can see that cluster 1 is associated with high GDPP and low child mortality, cluster 2 is associated with medium GDPP and medium child mortality, while cluster 3 shows low GDPP and high child mortality. (Please refer to table 1 in the annexes)

We will consider:

Cluster1: Developed countries

Cluster 2: Emerging countries

Cluster 3: Least developed countries

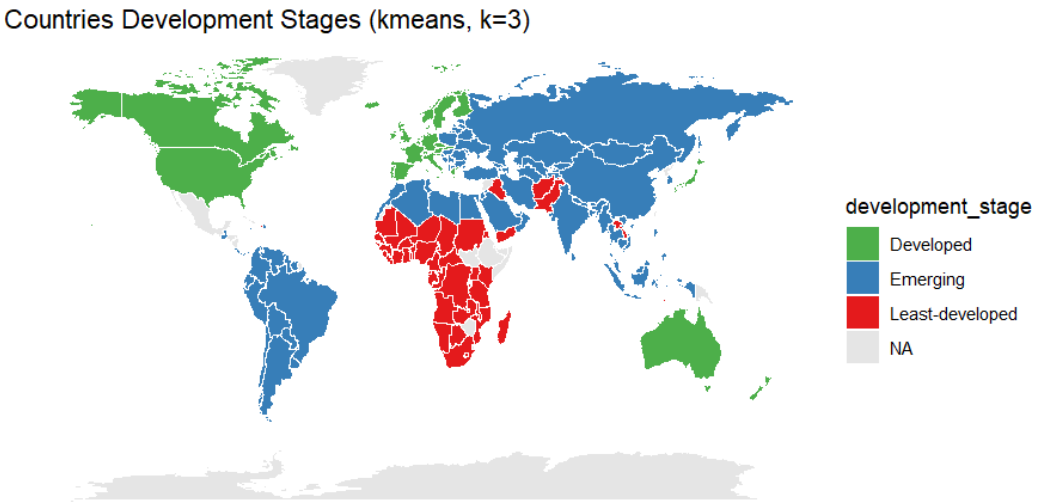


Figure 21: World map of Development Stages

Silhouette width:

Cluster	Size	Ave.sil.width
1	30	0.17
2	89	0.35
3	47	0.25

Table 3: Silhouette Width

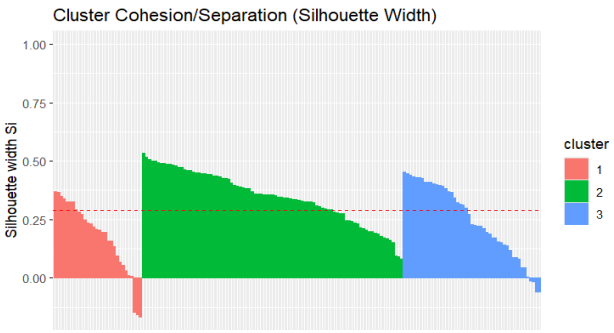


Figure 22: Silhouette width per cluster (kmeans, k=3)

The clustering results show average silhouette widths of 0.17 for Cluster 1, 0.35 for Cluster 2, and 0.25 for Cluster 3. Cluster 2 demonstrates moderate separation, while Clusters 1 and 3 indicate weaker structure, especially Cluster 1. Overall, the clustering provides some insights but suggests room for improvement in defining clearer group boundaries.

K-means clustering with k=4

cluster	median_gdpp	median_child_mort	n_countries
1	45350	4.2	24
2	46600	2.8	3
3	897	90.2	47
4	5430	17.3	92

Table 4: kmeans clusters with k=4

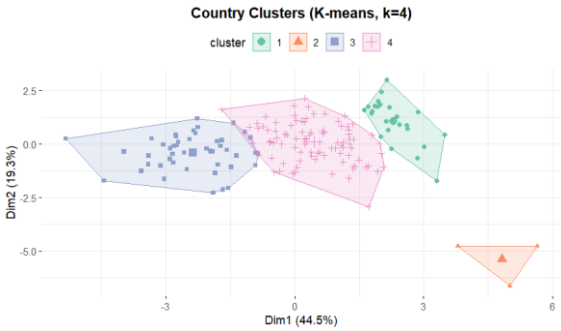


Figure 23: Country clusters k=4

K-means clustering with k=4 revealed significant imbalance across clusters. While Clusters 1 and 2 (high GDP, low child mortality) contain only 24 and 3 countries respectively, Clusters 3 and 4 include 47 and 92 countries, representing lower to middle-income nations with higher child mortality rates. This highlights major global inequalities in development and health outcomes. (please refer to table 2 in the Annexes section)

Kmeans on PCA

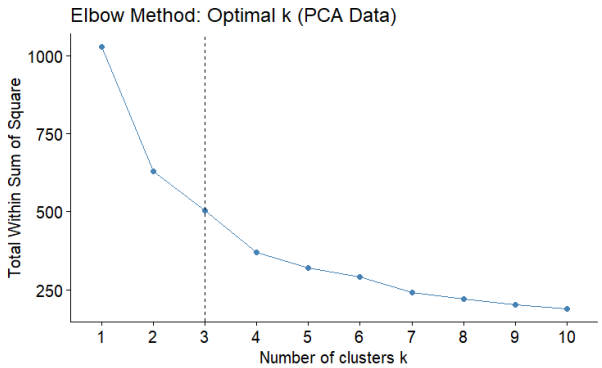


Figure 24: Elbow method for PCA data

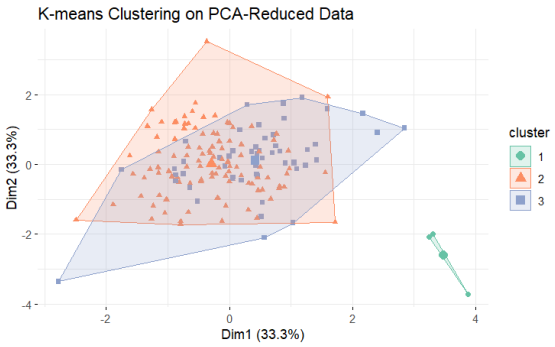


Figure 25: Country clusters for PCA data

While k-means clustering on the PCA-reduced data initially appeared promising, the 2D projection revealed significant overlap between two clusters. This suggests that:

Critical features for distinguishing these groups may reside in higher principal components

The 30%+ variance loss from dimensionality reduction obscured key development patterns

Original feature space clustering (shown in Figure 19) provides clearer separation

We prefer the k-means results that were performed on our original scaled data for better cluster differentiation and interpretability.

Hierarchical clustering

Hierarchical clustering is an unsupervised machine learning technique that builds a tree-like structure (dendrogram) to represent data groupings based on similarity. It does not require a pre-specified number of clusters and allows for a visual inspection of potential natural groupings.

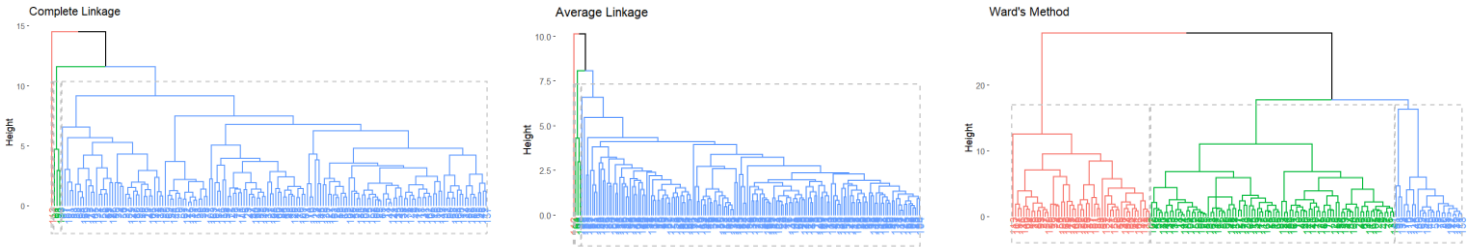


Figure 26: Hierarchical Clustering

In this project, hierarchical clustering was applied using complete, average, and Ward's linkage methods. Ward's method was selected for its better cluster compactness. Using k=3, we classified countries into "Least-developed," "Emerging," and "Developed" categories (please refer to table 3 in the annxes).

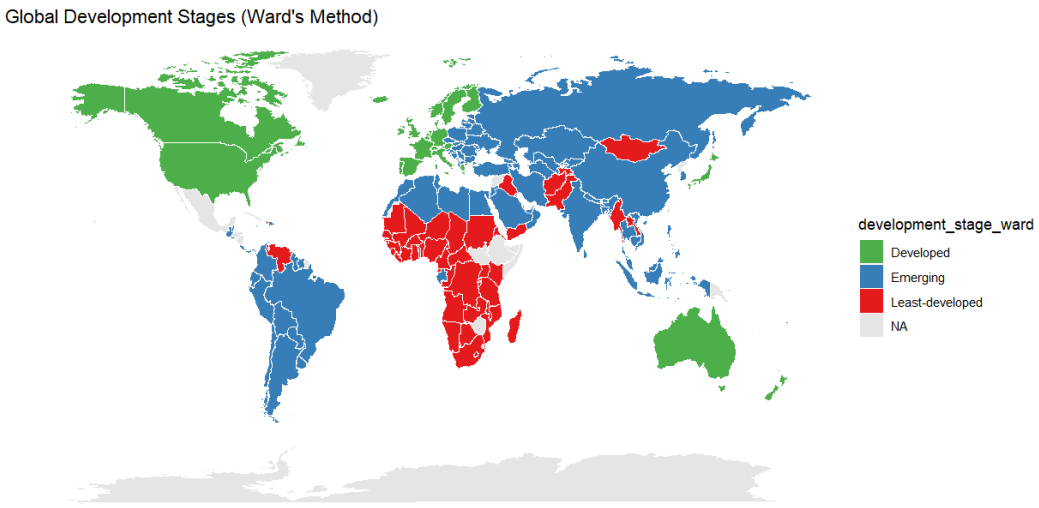


Figure 27: Clustering of countries by development stages with Ward Method

In order to evaluate the clustering quality, we used two metrics:

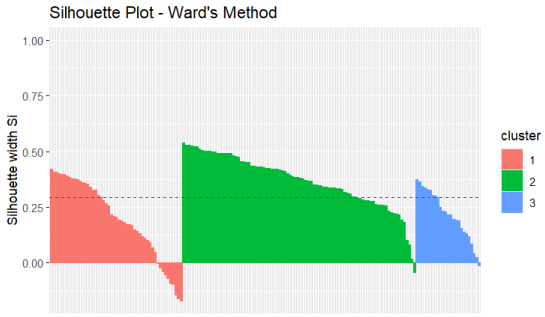


Figure 28: Silhouette width per cluster (hclust: Ward's method)

The average silhouette width was 0.29, suggesting that countries within the same cluster were relatively well-matched, while separation between clusters remained acceptable.

We also calculated the cophenetic correlation coefficient and we found a 0.532 correlation, indicating a moderate level of agreement between the original pairwise distances and the distances represented by the clustering dendrogram. This shows that the hierarchical structure captured a reasonable amount of the dataset's underlying similarity patterns.

Supervised Learning

Supervised learning is a fundamental machine learning technique where a model is trained on a dataset that includes both input features and known output labels. The goal is to learn the relationship between the features and the target variable so the model can accurately predict outcomes for new, unseen data. This approach is commonly used in classification problems, for this project, which focuses on predicting heart disease, we implemented and compared four supervised learning models: *Logistic Regression*, *Support Vector Machine (SVM)*, *Decision Tree*, and *Random Forest*.

Dataset

Cardiovascular diseases, particularly heart failure, are among the main causes of mortality nowadays. Early detection and accurate prediction of heart disease can significantly improve patient outcomes through timely intervention and treatment. In this project, we aim to develop predictive models that can identify the likelihood of a person having heart disease based on dataset selected from Kaggle, which contains 918 observations and 12 variables:

Variable Name	Type	Description
Age	Integer	Age of the patient [years]
Sex	Character	Sex of the patient [M: Male, F: Female]
ChestPainType	Character	Chest pain type [TA: Typical Angina, ATA: Atypical Angina, NAP: Non-Anginal Pain, ASY: Asymptomatic]
RestingBP	Integer	Resting blood pressure [mm Hg]
Cholesterol	Integer	Serum cholesterol [mm/dl]
FastingBS	Integer	Fasting blood sugar [1: if fastingbs > 120 mg/dl, 0: otherwise]
RestingECG	Character	Resting electrocardiogram results [Normal: Normal, ST: having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mv), LVH: showing probable or definite left ventricular hypertrophy by Estes' criteria]
MaxHR	Ineteger	Maximum heart rate achieved [Numeric value between 60 and 202]
ExerciseAngina	Character	Exercise-induced angina [Y: Yes, N: No]
Oldpeak	Numerical	Oldpeak = ST [Numeric value measured in depression]
ST_Slope	Character	The slope of the peak exercise ST segment [Up: upsloping, Flat: flat, Down: downsloping]
HeartDisease	Integer	Output class [1: heart disease, 0: Normal]

A. Exploratory Data Analysis (EDA)

I. Univariate Analysis

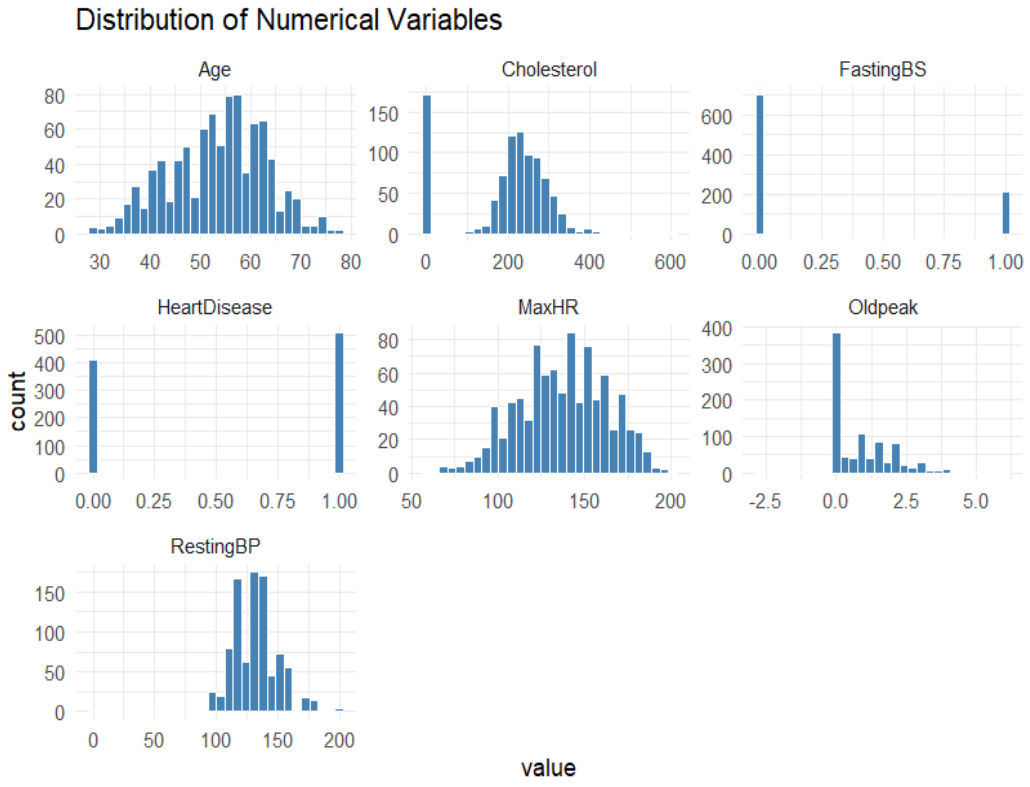


Figure 29: Distribution of Numerical Variables

According to our Dataset summary (see table 6 and 7 from annexes) and from the distributions above of our variables we can observe that:

Age and MaxHR variable are near to be normally distributed, Cholesterol is left skewed with many 0 variables that we should treat after checking for outliers present in our dataset.

RestingBP, Oldpeak appear to be right or left skewed with probability of the presence of outliers.

These distributions highlight need for further data exploration in order to check for outliers and possible unclean data.

Distributions of Heart Disease (target variable) and Sex

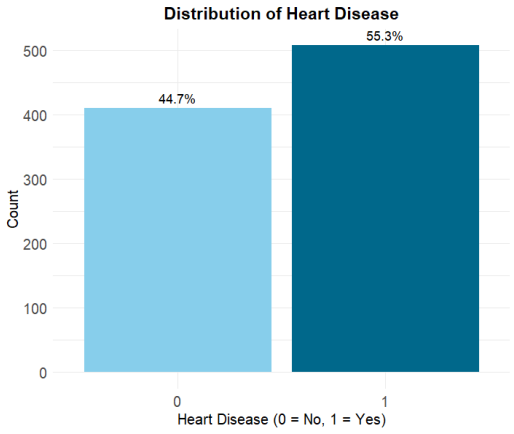


Figure 30: Distribution of Heart disease

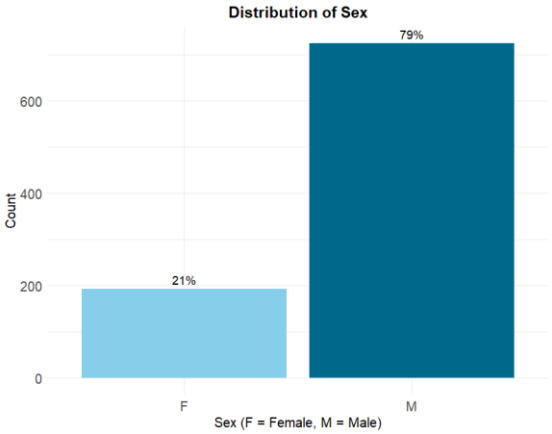


Figure 31: Distribution of Sex

According to figure 30, the number of patients with heart disease is more than the number of patients without a heart disease (the percentage of sick patients is 55.3 %), at first it may sound like the data is imbalanced, but this in fact can help us when predicting if a patient actually has a heart disease, also figure 31 reveals that the number males present in our dataset is way more than the number of females (males represent 79% of our data)

Boxplots for Numerical Variables

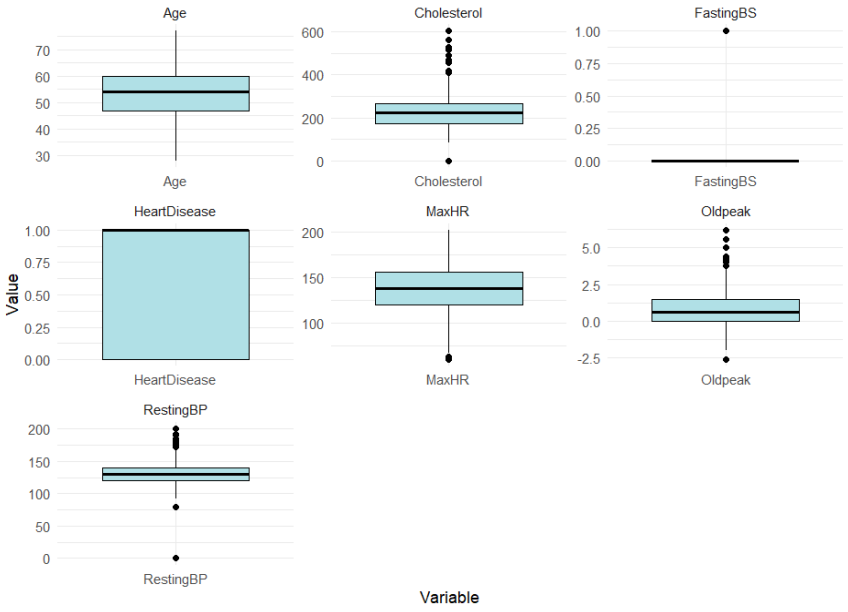


Figure 32: Boxplots for Numerical Variables

According to these boxplots we found that there's one row with RestingBP=0 which is an outlier because the RestingBP cannot be 0 so we have to remove this row.

In addition, after further analysis, we found 172 records with a cholesterol value of 0, which is medically implausible and likely indicates missing data; 152 of these cases also had heart disease, which could mislead our models if we don't correct them, as high cholesterol is typically associated with heart conditions. To address this, we replaced the zero values with NA and applied **K-Nearest Neighbors (KNN)** imputation to estimate realistic cholesterol values based on similar cases. This step will help us maintain our data quality and ensures our models learn from clinically valid inputs.

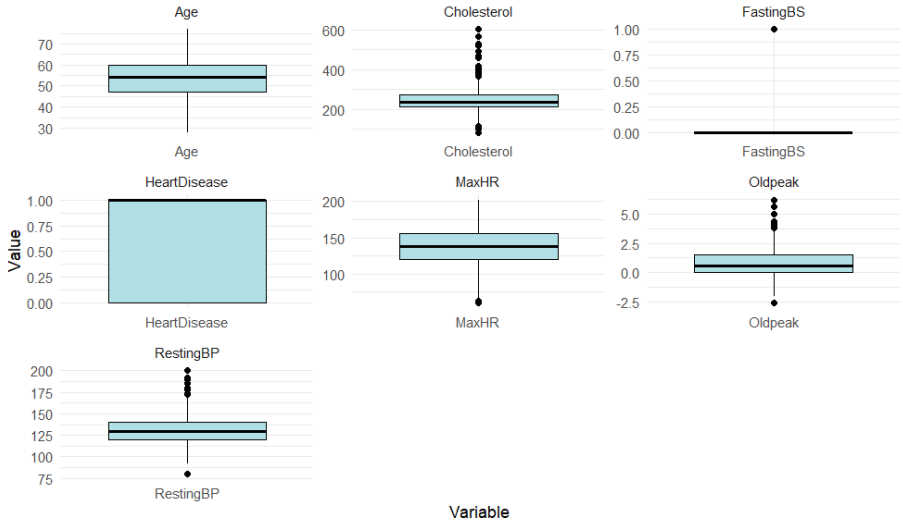


Figure 33: Boxplots for Numerical Variables after Handling Outliers

II. Bivariate Analysis

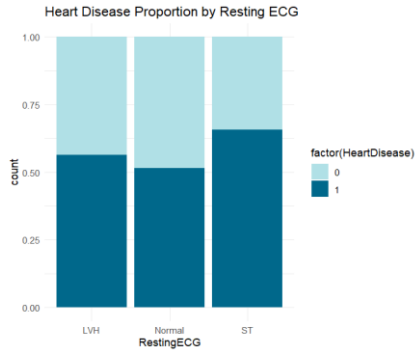


Figure 34: Heart Disease by Resting ECG

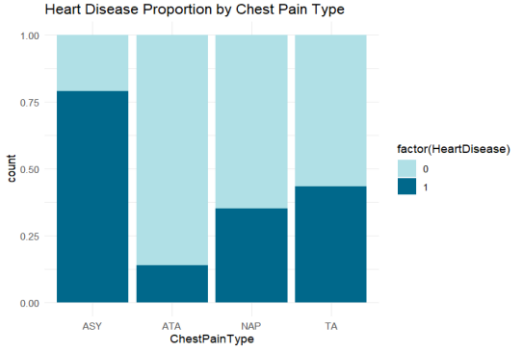


Figure 35: Heart Disease by Chest Pain Type

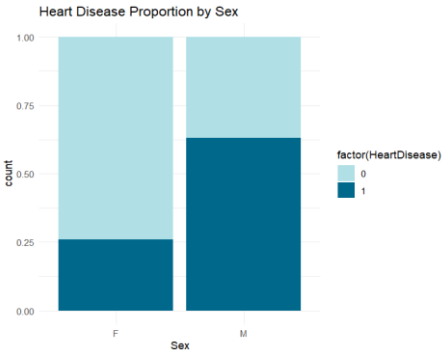


Figure 36: Heart Disease by Sex

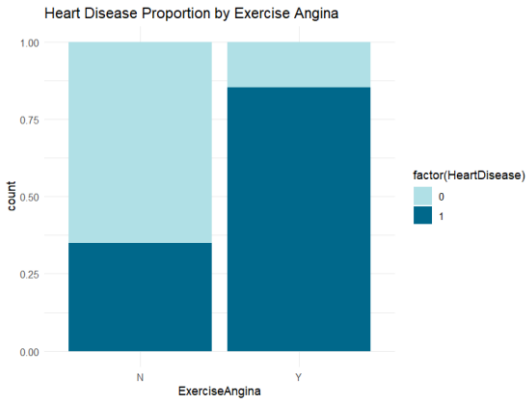


Figure 37: Heart Disease by Exercise Angina

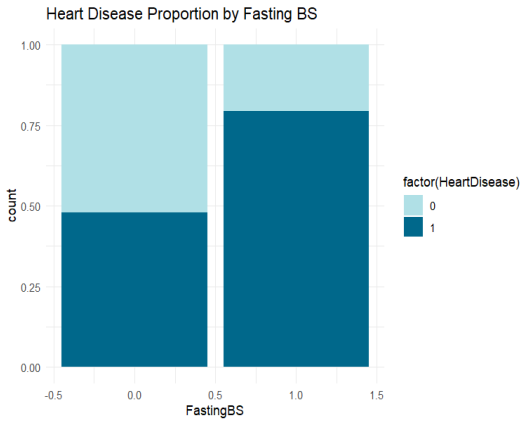


Figure 38: Heart Disease by Fasting BS

From these figures we can interpret that:

Most of Resting ECG with ST values correspond to patients who have a heart disease (Figure 34)

Chest Pain of type ASY coreesdpond to more heart diseased patients that the other types (nearly (79%) see figure 35.

Most of patients with heart disease are males, this can also be due to the fact that our data consist of 79% of males)

People with Exercise induced angina tend to have heart disease more than those who don't have.

Patients with Fasting Blood Sugar > 120 mg/dl tend to have a heart disease more.

III. Multivariate Analysis

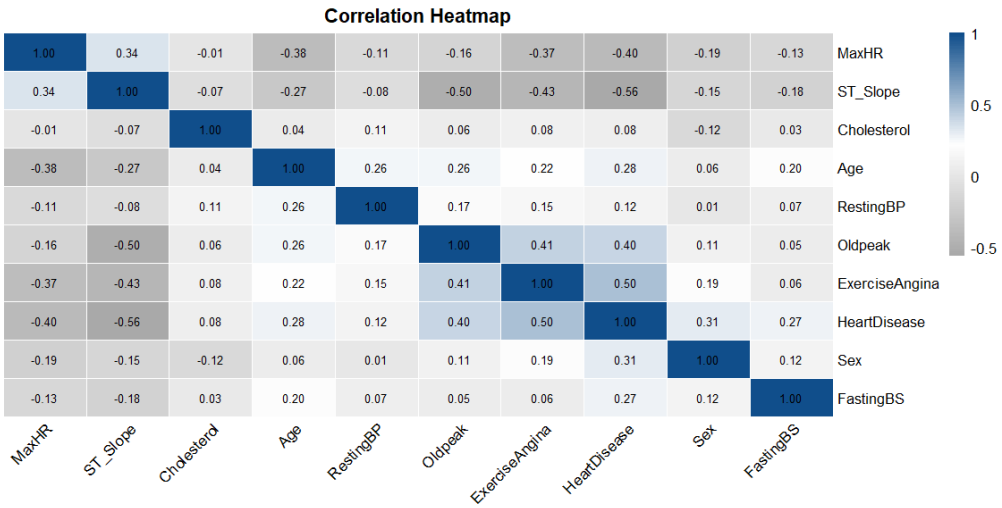


Figure 39: Correlation Heatmap

The correlation heatmap (figure 39) reveals several notable relationships between variables. The target variable, **HeartDisease**, shows a moderate positive correlation with **Oldpeak** (0.40), **ExerciseAngina** (0.50), and **Age** (0.28), while displaying a strong negative correlation with **ST_Slope** (-0.56) and **MaxHR** (-0.40). These relationships are consistent with medical expectations, where higher age, exercise-induced angina, and lower maximum heart rate tend to be associated with heart disease. Meanwhile, variables like **Cholesterol** and **RestingBP** have relatively weak correlations with HeartDisease, suggesting they may have less predictive value on their own. The heatmap also highlights internal correlations, such as the strong inverse relationship between **Oldpeak** and **ST_Slope** (-0.50), which may reflect overlapping clinical patterns.

B. Data Preprocessing

I. Train-Test Split

Tha main purpose of splitting our data is to evaluate the model's ability to generalize to new, unseen data, the dataset was divided into two subsets: a training set (80% of the data) and a test set (20% of the data). We used stratified sampling for the split to maintain the proportion of cases with and without heart disease. This ensures us that both the training and test sets have a similar distribution of target classes, which is crucial when the target variable is imbalanced.

II. Handling Categorical Variables

Categorical Variables: Variables like Sex, ExerciseAngina, ChestPainType, RestingECG, and ST_Slope are categorical in nature.

we first converted these categorical variables into factors, as most machine learning models (like logistic regression or decision tree) can handle them better when they are in factor form, we then transformed these factors into dummy (binary) variables. During This process we created a separate binary column for each level of a categorical variable (for example one for each type of chest pain). This allows the models to work with categorical data.

III. Feature Scaling

Numerical variables like Age, RestingBP, Cholesterol, MaxHR, and Oldpeak have different units and ranges (e.g., Cholesterol is typically in mg/dL, while Age is in years). If we do not scale them, some machine learning models may give too much importance to features with higher values or wider ranges, leading to poor model performance.

Hence, we standardized the numerical variables using z-score normalization. This transforms the data so that each feature has a mean of 0 and a standard deviation of 1, making them comparable across the different models.

IV. Cross validation

We used Cross-validation to assess the models' performance more robustly by training and testing the model multiple times on different data splits. This helps us avoid overfitting (when the model learns the training data too well and fails to generalize to new data).

The data was split into 10 equal folds, and the model was trained on 9 folds, while the remaining 1 fold was used for testing. This process was repeated 10 times, so each fold gets used for testing exactly once.

C. Models Implementation

I. Logistic Regression

Metric	Value	Interpretation
Accuracy	75.96%	Correctly classified 76% of cases
Sensitivity (Recall)	91.67%	Correctly identified 91.7% of true heart disease cases
Specificity	58.62%	Correctly identified 58.6% of healthy cases
False Positives	36	Incorrectly flagged 36 healthy patients
False Negatives	8	Missed 8 true heart disease cases

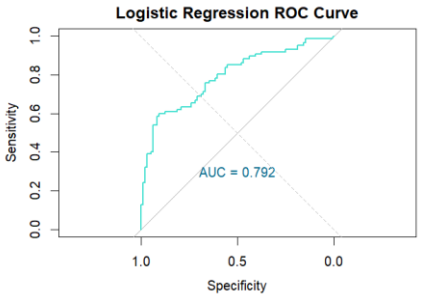


Figure 40: Logistic Regression ROC Curve

The Linear Regression model shows strong sensitivity (91.7%), missing only 8 true heart disease cases, which make it effective for initial screening where false negatives are critical. However, it has a moderate accuracy (76%) and a lower specificity (58.6%), it also produces 36 false positives. This suggests that it prioritizes safety over precision, and potentially requires secondary testing for flagged cases. This tradeoff is clinically acceptable for early detection but may increase unnecessary follow-ups.

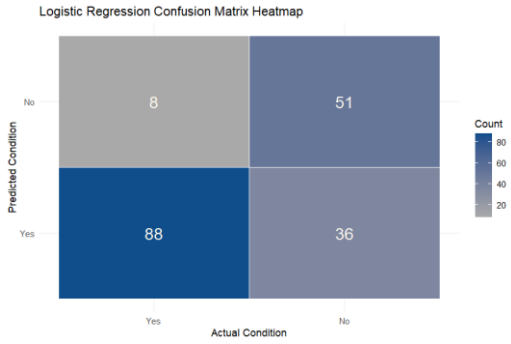


Figure 41: Logistic Regression Confusion Matrix Heatmap

II. Support Vector Machine (SVM)

Metric	Value	Interpretation
Accuracy	84.15%	Correctly classified 84.2% of cases
Sensitivity	89.58%	Missed only **10/183** true heart disease cases
Specificity	78.16%	Correctly identified **78.2% of healthy patients**
False Positives	19	**19/87** healthy patients incorrectly flagged
False Negatives	10	**10/96** high-risk cases missed

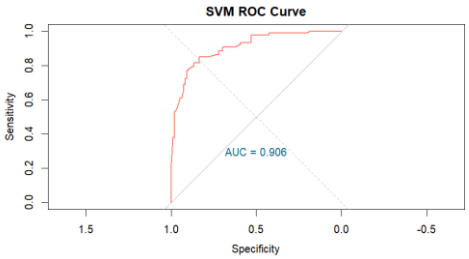


Figure 42: SVM ROC Curve

The polynomial SVM achieved 84.2% accuracy with strong sensitivity (89.6%, only 10 missed cases) and improved specificity (78.2%, 19 false alarms) versus logistic regression. Its excellent AUC (0.930) confirms effective non-linear pattern detection, making it ideal for initial screening where catching true cases outweighs unnecessary follow-ups. The 47% fewer false positives than logistic regression reduce clinical burden while maintaining safety.

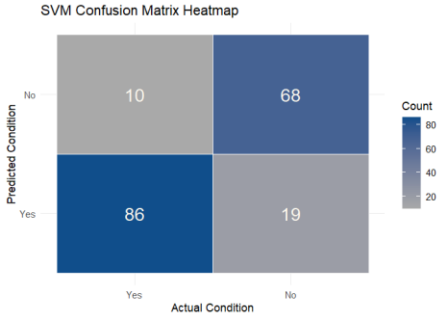


Figure 43: SVM Confusion Matrix Heatmap

III. Decision Tree

Metric	Value	Interpretation
Accuracy	77.60%	Correctly classified 77.6% of cases
Sensitivity	90.60%	Missed only **9/96** true heart disease cases
Specificity	63.20%	Correctly identified **63.2% of healthy patients**
False Positives	32	Incorrectly flagged 32 healthy patients
False Negatives	9	Missed 9 true heart disease cases

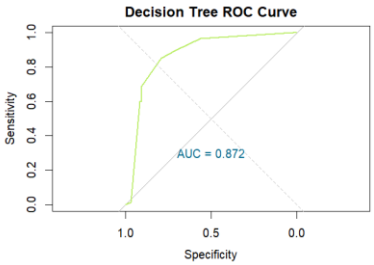


Figure 44: Decision Tree ROC Curve

The Decision Tree achieved 77.6% accuracy with high sensitivity (90.6%, missing only 9 true cases), making it effective for initial screening. Its specificity (63.2%) produces more false alarms (32 false positives) than SVM but fewer than logistic regression. The model's key strength is interpretability as the visualized tree shows ST segment changes and Chest Pain Type as primary decision points, aligning with clinical knowledge.

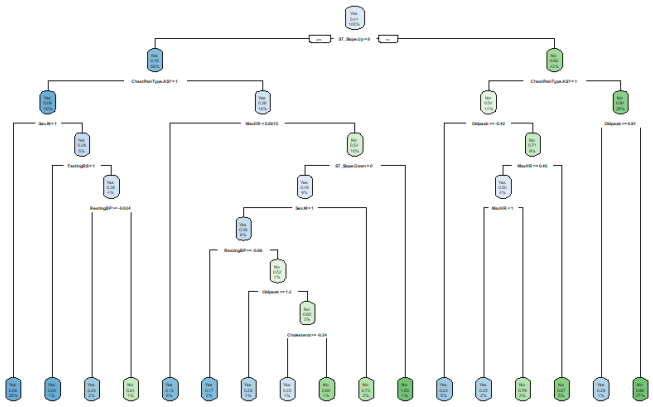


Figure 45: Decision Tree

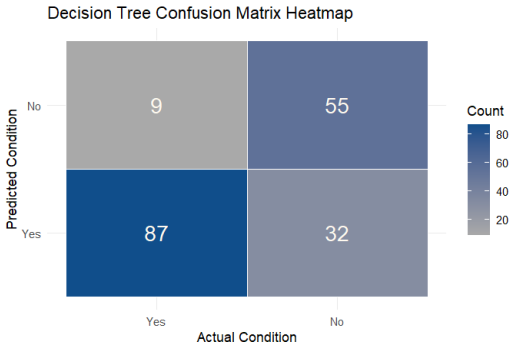


Figure 46: Decision Tree Confusion Matrix Heatmap

IV. Random Forest

Metric	Value	Interpretation
Accuracy	85.30%	Highest among all models
Sensitivity	90.60%	Missed only 9/96 true cases
Specificity	79.30%	Correctly identified 79.3% healthy patients
False Positives	18	Lowest FP count (vs. SVM's 19, DT's 32, LR's 36)
False Negatives	9	Missed 9 true heart disease cases

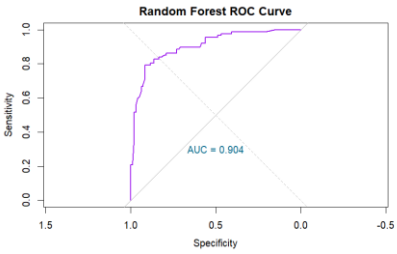


Figure 47: Random Forest ROC Curve

The Random Forest model demonstrated superior performance with 85.3% accuracy and excellent sensitivity (90.6%), missing only 9 true heart disease cases. Its high specificity (79.3%) resulted in just 18 false positives – the lowest among all tested models.

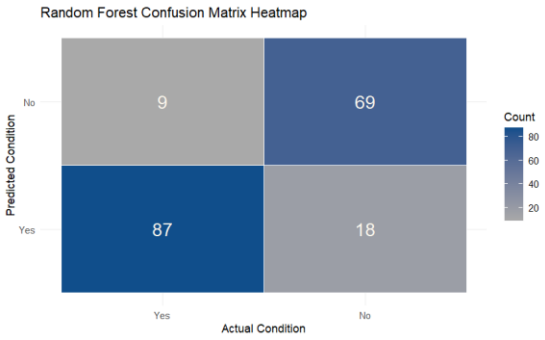
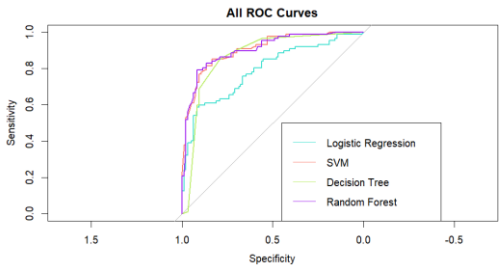


Figure 48: Random Forest Confusion Matrix Heatmap

V. Comparing The Models

Model	Accuracy	True Negatives	False Positives	False Negatives	True Positives
Logistic Regression	76.00%	88	36	8	51
SVM	84.20%	86	19	10	68
Decision Tree	77.60%	87	32	9	55
Random Forest	85.30%	87	18	9	69



Random Forest achieved the best performance (85.3% accuracy, 18 FPs), followed closely by SVM (84.2%, 19 FPs). While Decision Tree (77.6%, 32 FPs) offered interpretability and Logistic Regression (76.0%, 36 FPs) provided simplicity, all models maintained excellent sensitivity (>90%). Random Forest's optimal balance of high accuracy and low false positives makes it the top choice for clinical deployment, with SVM as a faster alternative.

Annexes

Unsupervised Learning

country	child_mort	exports	health	imports
Length:166	Min. : 2.60	Min. : 0.109	Min. : 1.810	Min. : 0.0659
Class :character	1st Qu.: 8.60	1st Qu.: 23.800	1st Qu.: 4.915	1st Qu.: 30.1000
Mode :character	Median : 19.50	Median : 35.200	Median : 6.270	Median : 43.3000
	Mean : 38.47	Mean : 41.146	Mean : 6.811	Mean : 46.9745
	3rd Qu.: 62.15	3rd Qu.: 51.375	3rd Qu.: 8.625	3rd Qu.: 58.8250
	Max. :208.00	Max. :200.000	Max. :17.900	Max. :174.0000

income	inflation	life_expec	total_fer	gdpp
Min. : 609	Min. : -4.210	Min. :32.10	Min. :1.150	Min. : 231
1st Qu.: 3348	1st Qu.: 1.857	1st Qu.:65.30	1st Qu.:1.792	1st Qu.: 1320
Median : 9950	Median : 5.415	Median :73.10	Median :2.395	Median : 4635
Mean : 17070	Mean : 7.818	Mean :70.49	Mean :2.947	Mean : 12858
3rd Qu.: 22600	3rd Qu.: 10.825	3rd Qu.:76.78	3rd Qu.:3.895	3rd Qu.: 13500
Max. :125000	Max. :104.000	Max. :82.80	Max. :7.490	Max. :105000

Table 1: Country dataset Summary

```
colSums(is.na(df))
```

country	child_mort	exports	health	imports	income	inflation	life_expec	total_fer	gdpp
0	0	0	0	0	0	0	0	0	0

Table 2: Country Dataset NA values

Cluster Characteristics (k=3)

Cluster	Countries	Child Mort.	Exports	Health	Imports	Inflation	Life Expect.	Fertility	GDPP
1	**30**	4.43	58.30	9.61	53.39	1.45	80.51	1.69	44920.00
2	**89**	21.17	41.70	6.09	47.27	7.75	73.08	2.28	7825.42
3	**47**	92.96	29.15	6.39	42.32	12.02	59.19	5.01	1922.38

Table 3: Mean values for k-means clustering with k=3.

Cluster Characteristics (k=4)

Cluster	Countries	Child Mort.	Exports	Health	Imports	Inflation	Life Expect.	Fertility	GDPP
1	24	4.4	42.9	10.2	38.9	1.5	80.8	1.8	46154.2
2	3	4.1	176.0	6.8	156.7	2.5	81.4	1.4	57566.7
3	47	93.0	29.2	6.4	42.3	12.0	59.2	5.0	1922.4
4	92	20.6	42.4	6.1	47.9	7.5	73.2	2.3	8300.7

Table 4: Mean values for k-means clustering with k=4.

Average Variable Values per Cluster (Ward's Method)

cluster	child_mort	exports	health	imports	inflation	life_expec	total_fer	gdpp
1	1.24	-0.48	-0.13	-0.17	0.48	-1.18	1.23	-0.59
2	-0.46	0.11	-0.27	0.03	-0.10	0.34	-0.47	-0.19
3	-0.85	0.58	1.22	0.23	-0.60	1.18	-0.81	1.89

Table 5: Mean values for Hierarchical clustering (Ward's method).

Supervised Learning

Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS
Min. :28.00	Length:918	Length:918	Min. : 0.0	Min. : 0.0	Min. :0.0000
1st Qu.:47.00	Class :character	Class :character	1st Qu.:120.0	1st Qu.:173.2	1st Qu.:0.0000
Median :54.00	Mode :character	Mode :character	Median :130.0	Median :223.0	Median :0.0000
Mean :53.51			Mean :132.4	Mean :198.8	Mean :0.2331
3rd Qu.:60.00			3rd Qu.:140.0	3rd Qu.:267.0	3rd Qu.:0.0000
Max. :77.00			Max. :200.0	Max. :603.0	Max. :1.0000

RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
Length:918	Min. : 60.0	Length:918	Min. :-2.6000	Length:918	Min. :0.0000
Class :character	1st Qu.:120.0	Class :character	1st Qu.: 0.0000	Class :character	1st Qu.:0.0000
Mode :character	Median :138.0	Mode :character	Median : 0.6000	Mode :character	Median :1.0000
	Mean :136.8		Mean : 0.8874		Mean :0.5534
	3rd Qu.:156.0		3rd Qu.: 1.5000		3rd Qu.:1.0000
	Max. :202.0		Max. : 6.2000		Max. :1.0000

Table 6: Heart Dataset Summary

Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG
0	0	0	0	0	0	0

MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	0	0	0	0

Table 7: Heart Dataset NA Values